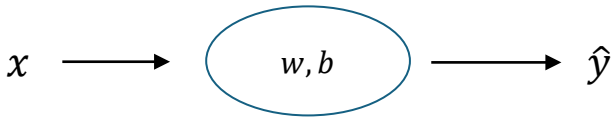
Perceptron vector notation



Input

$X_{(n,m)}$
$y_{(n,)}$
$n - number\ of\ data\ points$
$m - dimension\ of\ data\ point$

Parameters:

$w(m,)$
$b(1,)$

Prediction:

$$\underset{(n,m)}{X} \cdot \underset{(m,1)}{w_1} + \underset{(1)}{b_1} = \underset{(n,1)}{z}$$

$$\hat{y} = \begin{cases} z \geq 0 & 1 \\ z < 0 & 0 \end{cases}$$

Update:

$$update = \eta(y - \hat{y})$$
$$(n,) \quad 1(n,)(n,)$$

$$w_{new} = w_{old} + \sum_{i=1}^{n}(update \circ X)$$
$$(m,) \quad (m,) \quad (n,)(n,m)$$

$$b_{new} = b_{old} + \sum_{i=1}^{n}(update)$$
$$(1,) \quad (1,) \quad (n,)$$